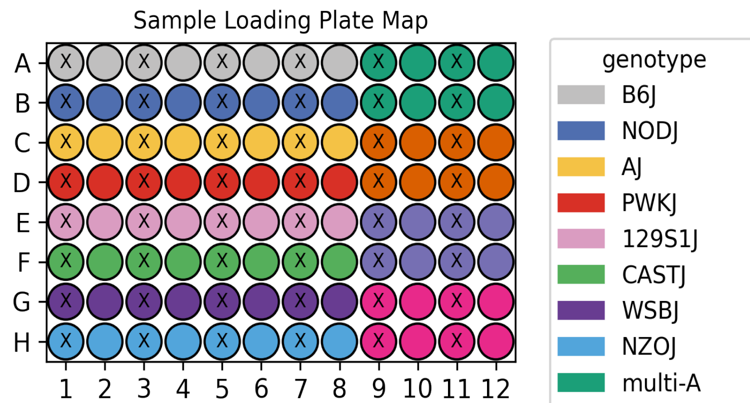
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_008
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	GonadsFemale GonadsMale Adrenal
Subpools	9
Exome capture subpools	0
Expected nuclei	1,000,000



1. Sample multiplexing

GonadsMale and GonadsFemale were the main tissues on the plate while Adrenal was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ

2. Alignment

A total of 11,477,408,600 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_2	1,315,273,634	625,751,760	0.19	772,468,467	58.7
Subpool_3	1,184,655,984	541,770,026	0.252	724,479,295	61.2
Subpool_4	1,363,606,649	600,217,741	0.267	819,154,750	60.1
Subpool_5	1,228,834,464	539,676,433	0.263	732,735,735	59.6
Subpool_6	1,361,953,811	671,728,703	0.211	851,475,785	62.5
Subpool_7	1,344,200,288	636,856,891	0.245	843,105,553	62.7
Subpool_8	1,175,974,432	580,268,014	0.213	737,673,044	62.7
Subpool_9	1,261,891,071	607,044,903	0.237	795,565,352	63.0
Subpool_10	1,241,018,267	600,721,238	0.231	781,050,407	62.9

3. Cell recovery

A total of 2,020,386 nuclei across all 9 subpools have >=200 UMIs and 738,842 nuclei have >= 500 UMIs.

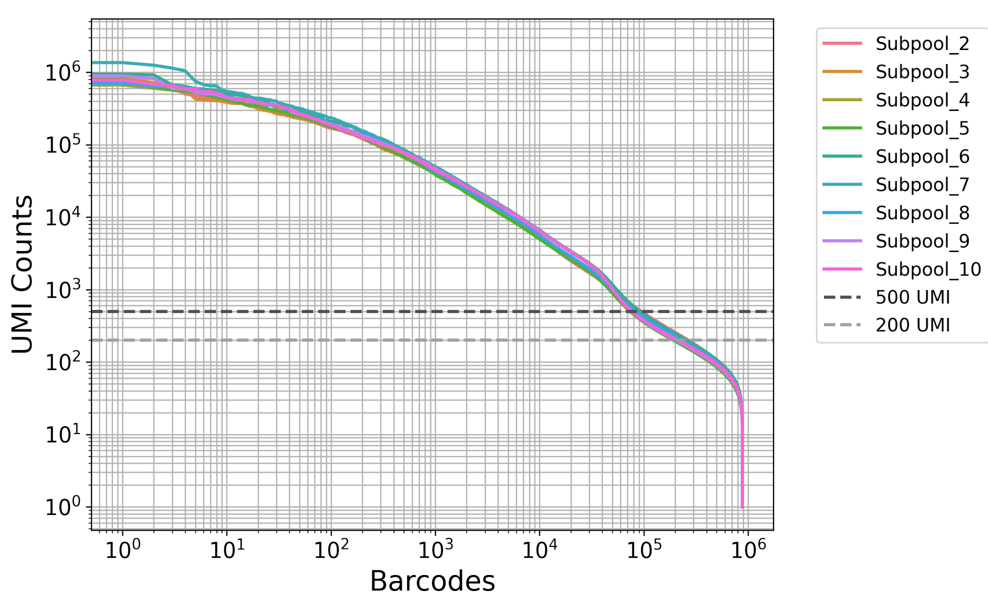


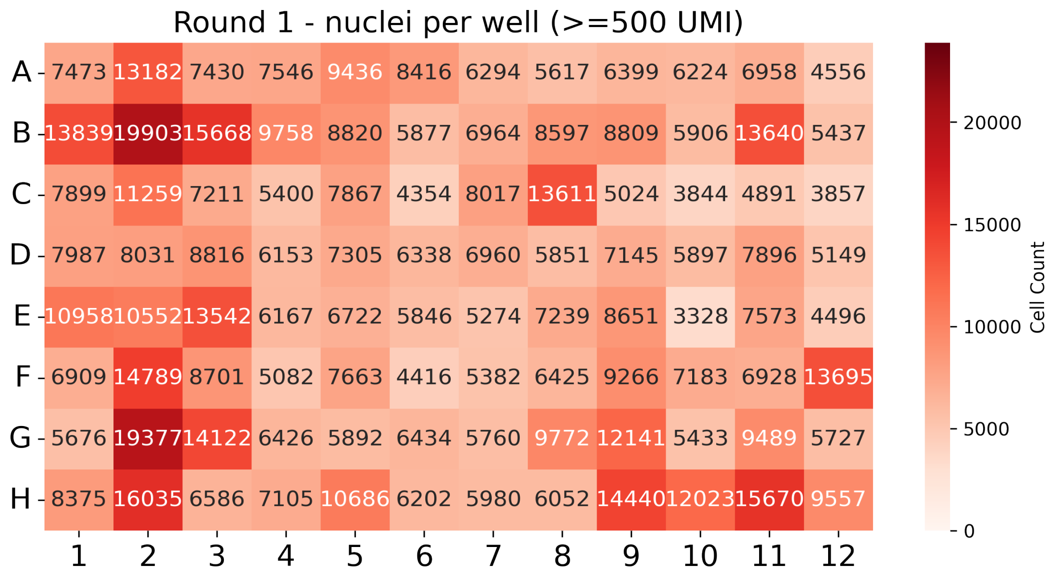
Table 3: QC stats for nuclei >=200 UMI

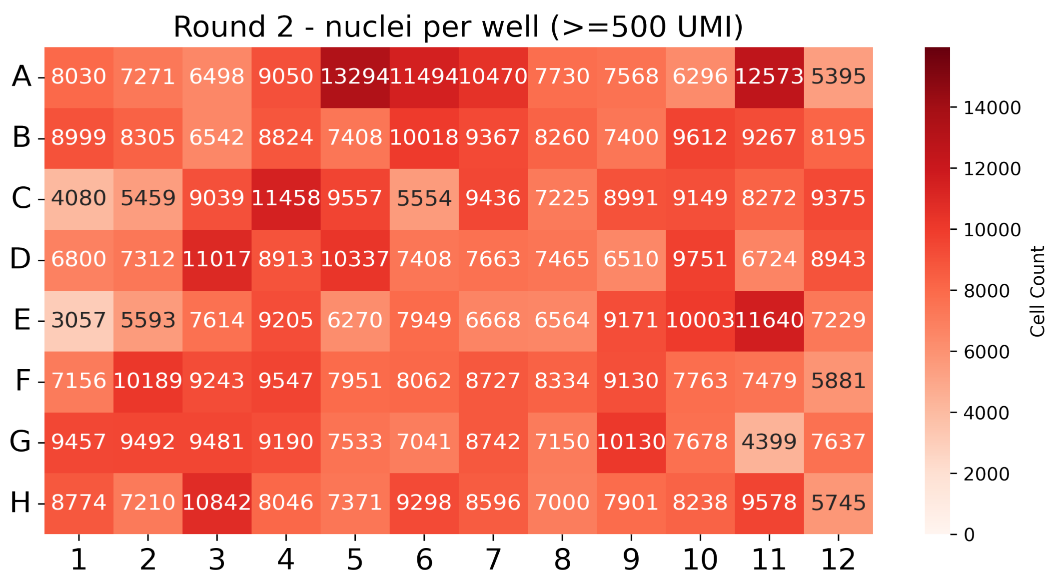
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_2	266,766	362	1,458.3	324	669.2
Subpool_3	213,989	365	1,587.9	325	717.3
Subpool_4	221,604	359	1,689.7	320	729.8
Subpool_5	192,909	368	1,728.7	328	749.3
Subpool_6	258,118	357	1,718.4	319	728.9
Subpool_7	219,046	363	1,939.2	324	779.1
Subpool_8	231,709	359	1,641.4	320	717.4
Subpool_9	203,939	361	1,953.5	322	803.7
Subpool_10	212,306	351	1,862.5	314	777.7

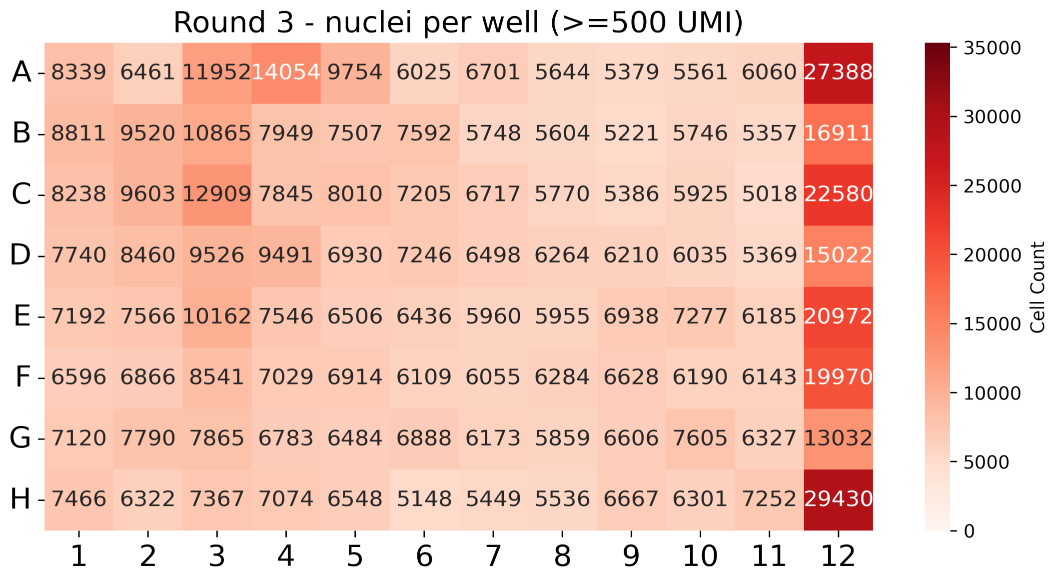
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_2	94,205	1,073	3,580.6	813	1,401.4
Subpool_3	79,623	1,299	3,769.3	919	1,479.8
Subpool_4	80,987	1,356	4,112.9	954	1,539.3
Subpool_5	74,120	1,385	4,029.3	957	1,528.7
Subpool_6	91,724	1,340	4,298.3	968	1,569.1
Subpool_7	81,570	1,507	4,711.0	1,034	1,646.1
Subpool_8	83,351	1,307	4,036.9	936	1,521.4
Subpool_9	76,552	1,634	4,718.0	1,094	1,704.3
Subpool_10	76,710	1,653	4,640.3	1,102	1,690.2

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.


Round 1 - nuclei per well (>=500 UMI)
A 7473 13182 7430 7546 9436 8416 6294 5617 6399 6224 6958 4556
B 13839 19903 15668 9758 8820 5877 6964 8597 8809 5906 13640 5437
C 7899 11259 7211 5400 7867 4354 8017 13611 5024 3844 4891 3857
D 7987 8031 8816 6153 7305 6338 6960 5851 7145 5897 7896 5149
E 10958 10552 13542 6167 6722 5846 5274 7239 8651 3328 7573 4496
F 6909 14789 8701 5082 7663 4416 5382 6425 9266 7183 6928 13695
G 5676 19377 14122 6426 5892 6434 5760 9772 12141 5433 9489 5727
H 8375 16035 6586 7105 10686 6202 5980 6052 14440 12023 15670 9557
1 2 3 4 5 6 7 8 9 10 11 12
20000
15000
10000
5000
0
Cell Count


Round 2 - nuclei per well (>=500 UMI)
A 8030 7271 6498 9050 13294 11494 10470 7730 7568 6296 12573 5395
B 8999 8305 6542 8824 7408 10018 9367 8260 7400 9612 9267 8195
C 4080 5459 9039 11458 9557 5554 9436 7225 8991 9149 8272 9375
D 6800 7312 11017 8913 10337 7408 7663 7465 6510 9751 6724 8943
E 3057 5593 7614 9205 6270 7949 6668 6564 9171 10003 11640 7229
F 7156 10189 9243 9547 7951 8062 8727 8334 9130 7763 7479 5881
G 9457 9492 9481 9190 7533 7041 8742 7150 10130 7678 4399 7637
H 8774 7210 10842 8046 7371 9298 8596 7000 7901 8238 9578 5745
1 2 3 4 5 6 7 8 9 10 11 12
14000
12000
10000
8000
6000
4000
2000
0
Cell Count



4. Background removal

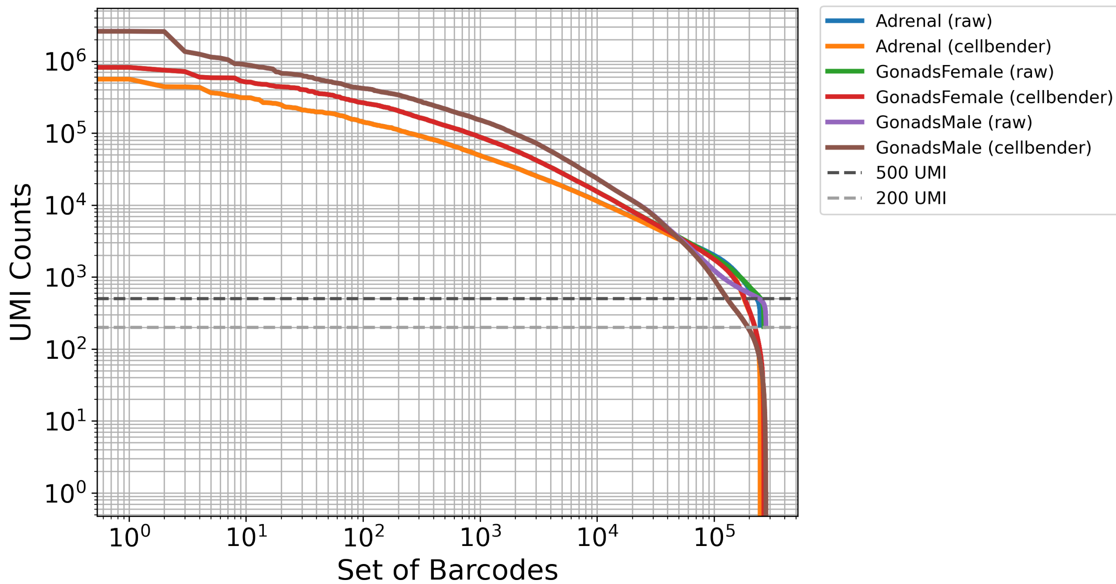
Table 5: Cellbender parameters

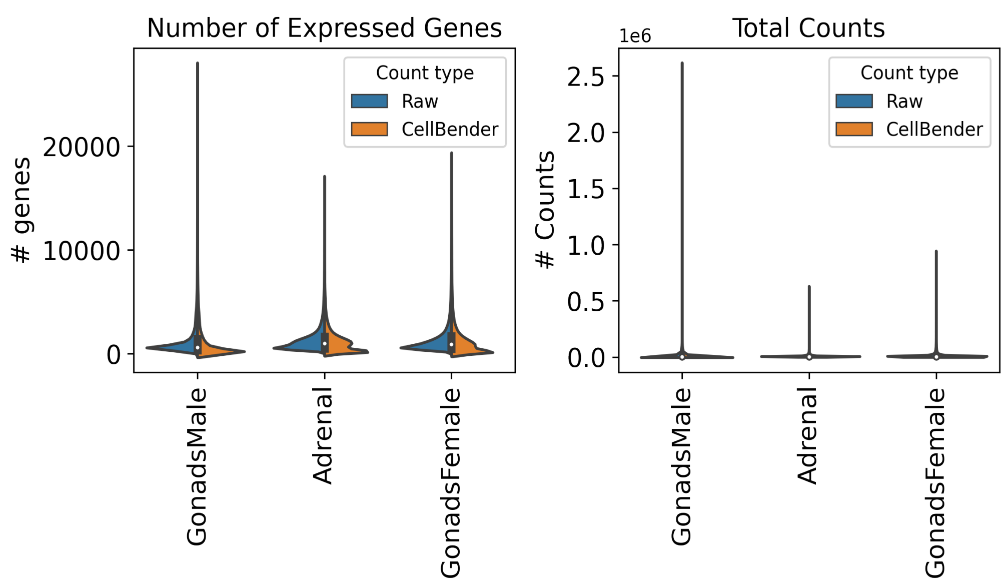
Subpool	droplets_included	learning_rate	expected_cells
Subpool_2	150000	5e-05	67000
Subpool_3	200000	5e-05	67000
Subpool_4	200000	5e-05	67000
Subpool_5	200000	5e-05	67000
Subpool_6	200000	5e-05	67000
Subpool_7	200000	5e-05	67000
Subpool_8	150000	5e-05	67000
Subpool_9	200000	5e-05	67000
Subpool_10	200000	5e-05	67000

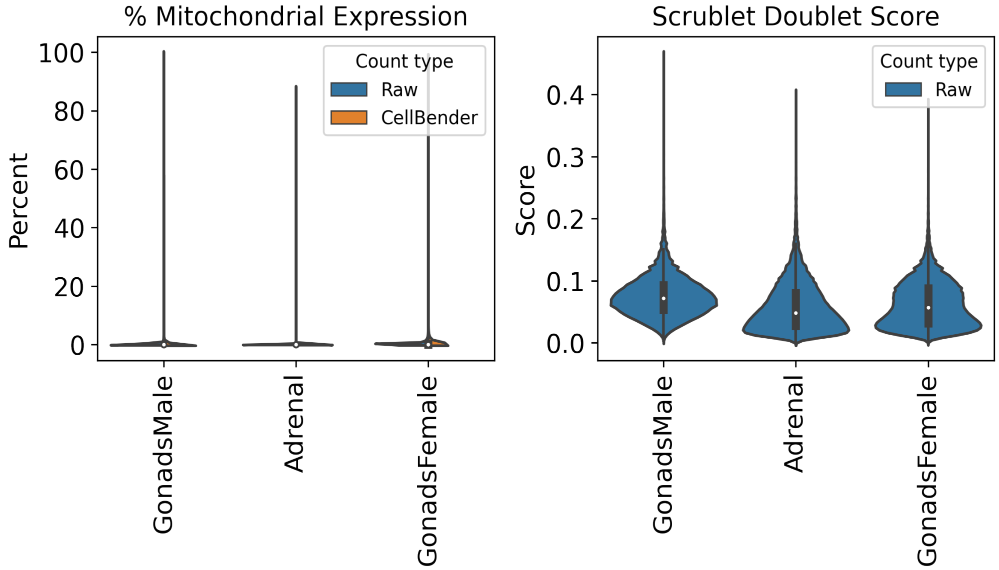
Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_2	8.3	89,765	3,420.6
Subpool_3	6.4	85,934	3,298.9
Subpool_4	6.0	84,128	3,735.1
Subpool_5	5.8	81,886	3,476.4
Subpool_6	6.2	91,008	4,057.1

Subpool_7	5.6	89,750	4,080.9
Subpool_8	7.0	81,484	3,828.9
Subpool_9	5.3	81,978	4,201.6
Subpool_10	6.8	105,325	3,251.8







5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM4567NUXK	016_B6J_10F_10	16	R	A1
CATTCCTA	IGVFSM4567NUXK	016_B6J_10F_10	16	T	A1
CATCATCC	IGVFSM3537SUZG	016_B6J_10F_11	16	R	A1
CATTCCTA	IGVFSM3537SUZG	016_B6J_10F_11	16	T	A1
CTGCTTTG	IGVFSM6618MHBV	017_B6J_10M_10	16	R	A2
CTTCATCA	IGVFSM6618MHBV	017_B6J_10M_10	16	T	A2
CTGCTTTG	IGVFSM1349YMMU	017_B6J_10M_11	16	R	A2
CTTCATCA	IGVFSM1349YMMU	017_B6J_10M_11	16	T	A2
CTAAGGGA	IGVFSM7587IQZO	018_B6J_10F_10	16	R	A3
CCTATATC	IGVFSM7587IQZO	018_B6J_10F_10	16	T	A3

CTAAGGGA	IGVFSM2910IGST	018_B6J_10F_11	16	R	A3
CCTATATC	IGVFSM2910IGST	018_B6J_10F_11	16	T	A3
GCTTATAG	IGVFSM4552FSZN	019_B6J_10M_10	16	R	A4
ACATTTAC	IGVFSM4552FSZN	019_B6J_10M_10	16	T	A4
GCTTATAG	IGVFSM4635AIZP	019_B6J_10M_11	16	R	A4
ACATTTAC	IGVFSM4635AIZP	019_B6J_10M_11	16	T	A4
TCTGATCC	IGVFSM6410RGFM	020_B6J_10F_10	16	R	A5
ACTTAGCT	IGVFSM6410RGFM	020_B6J_10F_10	16	T	A5
TCTGATCC	IGVFSM8699AZSF	020_B6J_10F_11	16	R	A5
ACTTAGCT	IGVFSM8699AZSF	020_B6J_10F_11	16	T	A5
TCTCTTGG	IGVFSM8837MYHY	021_B6J_10M_10	16	R	A6
CCAATTCT	IGVFSM8837MYHY	021_B6J_10M_10	16	T	A6
TCTCTTGG	IGVFSM5614XQUG	021_B6J_10M_11	16	R	A6
CCAATTCT	IGVFSM5614XQUG	021_B6J_10M_11	16	T	A6
CAATTTCC	IGVFSM1095OHOH	024_B6J_10F_10	16	R	A7
GCCTATCT	IGVFSM1095OHOH	024_B6J_10F_10	16	T	A7
CAATTTCC	IGVFSM4730CIUG	024_B6J_10F_11	16	R	A7
GCCTATCT	IGVFSM4730CIUG	024_B6J_10F_11	16	T	A7
AGTCTCTT	IGVFSM3397IFNI	025_B6J_10M_10	16	R	A8
ATGCTGCT	IGVFSM3397IFNI	025_B6J_10M_10	16	T	A8
AGTCTCTT	IGVFSM4481IPHX	025_B6J_10M_11	16	R	A8
ATGCTGCT	IGVFSM4481IPHX	025_B6J_10M_11	16	T	A8
TGCTGCTC	IGVFSM9731QOVH	016_B6J_10F_08	16	R	A9
CATTTACA	IGVFSM9731QOVH	016_B6J_10F_08	16	T	A9
TGCTGCTC	IGVFSM1981FBYZ	066_NODJ_10F_08	16	R	A9
CATTTACA	IGVFSM1981FBYZ	066_NODJ_10F_08	16	T	A9
GTATTTCC	IGVFSM7981FEKQ	017_B6J_10M_08	16	R	A10
ACTCGTAA	IGVFSM7981FEKQ	017_B6J_10M_08	16	T	A10
GTATTTCC	IGVFSM0149TNQV	067_NODJ_10M_08	16	R	A10
ACTCGTAA	IGVFSM0149TNQV	067_NODJ_10M_08	16	T	A10
TTCCTGTG	IGVFSM7273NNBM	018_B6J_10F_08	16	R	A11
CCTTTGCA	IGVFSM7273NNBM	018_B6J_10F_08	16	T	A11
TTCCTGTG	IGVFSM2286NVNW	068_NODJ_10F_08	16	R	A11
CCTTTGCA	IGVFSM2286NVNW	068_NODJ_10F_08	16	T	A11
GCTGCTTC	IGVFSM2795NYTC	019_B6J_10M_08	16	R	A12

ACTCCTGC	IGVFSM2795NYTC	019_B6J_10M_08	16	T	A12
GCTGCTTC	IGVFSM0256YAXR	069_NODJ_10M_08	16	R	A12
ACTCCTGC	IGVFSM0256YAXR	069_NODJ_10M_08	16	T	A12
TATGTGTC	IGVFSM9963SJEX	066_NODJ_10F_10	16	R	B1
ATTTGGCA	IGVFSM9963SJEX	066_NODJ_10F_10	16	T	B1
TATGTGTC	IGVFSM3417ZEGZ	066_NODJ_10F_11	16	R	B1
ATTTGGCA	IGVFSM3417ZEGZ	066_NODJ_10F_11	16	T	B1
CAATTCTC	IGVFSM8361NBRE	067_NODJ_10M_10	16	R	B2
TTATTCTG	IGVFSM8361NBRE	067_NODJ_10M_10	16	T	B2
CAATTCTC	IGVFSM5322PFJO	067_NODJ_10M_11	16	R	B2
TTATTCTG	IGVFSM5322PFJO	067_NODJ_10M_11	16	T	B2
TGGTCTCC	IGVFSM5649EBMW	068_NODJ_10F_10	16	R	B3
TCATGCTC	IGVFSM5649EBMW	068_NODJ_10F_10	16	T	B3
TGGTCTCC	IGVFSM3199IWJM	068_NODJ_10F_11	16	R	B3
TCATGCTC	IGVFSM3199IWJM	068_NODJ_10F_11	16	T	B3
GCTCTTTA	IGVFSM0379DQEB	069_NODJ_10M_10	16	R	B4
CATACTTC	IGVFSM0379DQEB	069_NODJ_10M_10	16	T	B4
GCTCTTTA	IGVFSM5988JHKG	069_NODJ_10M_11	16	R	B4
CATACTTC	IGVFSM5988JHKG	069_NODJ_10M_11	16	T	B4
GCTGCATG	IGVFSM6896ZGQR	070_NODJ_10F_10	16	R	B5
CCGTTCTA	IGVFSM6896ZGQR	070_NODJ_10F_10	16	T	B5
GCTGCATG	IGVFSM7472UIHU	070_NODJ_10F_11	16	R	B5
CCGTTCTA	IGVFSM7472UIHU	070_NODJ_10F_11	16	T	B5
ACTCATTT	IGVFSM2896HLSZ	071_NODJ_10M_10	16	R	B6
GCTTCATA	IGVFSM2896HLSZ	071_NODJ_10M_10	16	T	B6
ACTCATTT	IGVFSM3736YYMT	071_NODJ_10M_11	16	R	B6
GCTTCATA	IGVFSM3736YYMT	071_NODJ_10M_11	16	T	B6
AGTCTTGG	IGVFSM0625BIBT	072_NODJ_10F_10	16	R	B7
CTCTGTGC	IGVFSM0625BIBT	072_NODJ_10F_10	16	T	B7
AGTCTTGG	IGVFSM7300VPTD	072_NODJ_10F_11	16	R	B7
CTCTGTGC	IGVFSM7300VPTD	072_NODJ_10F_11	16	T	B7
GGTTCTTC	IGVFSM4303UBDW	073_NODJ_10M_10	16	R	B8
CCCTTATA	IGVFSM4303UBDW	073_NODJ_10M_10	16	T	B8
GGTTCTTC	IGVFSM8768SXWV	073_NODJ_10M_11	16	R	B8
CCCTTATA	IGVFSM8768SXWV	073_NODJ_10M_11	16	T	B8

TCATGTTG	IGVFSM2926ETSH	020_B6J_10F_08	16	R	B9
ACTGCTCT	IGVFSM2926ETSH	020_B6J_10F_08	16	T	B9
TCATGTTG	IGVFSM6603NGXT	070_NODJ_10F_08	16	R	B9
ACTGCTCT	IGVFSM6603NGXT	070_NODJ_10F_08	16	T	B9
ATTTTGCC	IGVFSM4409GRZD	021_B6J_10M_08	16	R	B10
CTCTAATC	IGVFSM4409GRZD	021_B6J_10M_08	16	T	B10
ATTTTGCC	IGVFSM2251OZKQ	071_NODJ_10M_08	16	R	B10
CTCTAATC	IGVFSM2251OZKQ	071_NODJ_10M_08	16	T	B10
CTTCTGTA	IGVFSM6796IKTY	024_B6J_10F_08	16	R	B11
ACCCTTGC	IGVFSM6796IKTY	024_B6J_10F_08	16	T	B11
CTTCTGTA	IGVFSM0170ECBA	072_NODJ_10F_08	16	R	B11
ACCCTTGC	IGVFSM0170ECBA	072_NODJ_10F_08	16	T	B11
GTCCATCT	IGVFSM4127LBLM	025_B6J_10M_08	16	R	B12
ATCTTAGG	IGVFSM4127LBLM	025_B6J_10M_08	16	T	B12
GTCCATCT	IGVFSM5573LEDH	073_NODJ_10M_08	16	R	B12
ATCTTAGG	IGVFSM5573LEDH	073_NODJ_10M_08	16	T	B12
GCTATCTC	IGVFSM7508KQHS	026_AJ_10F_10	16	R	C1
CATGTCTC	IGVFSM7508KQHS	026_AJ_10F_10	16	T	C1
GCTATCTC	IGVFSM9463TXUH	026_AJ_10F_11	16	R	C1
CATGTCTC	IGVFSM9463TXUH	026_AJ_10F_11	16	T	C1
TAGTTTCC	IGVFSM6129LTHE	029_AJ_10M_10	16	R	C2
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TAGTTTCC	IGVFSM9563RZBJ	029_AJ_10M_11	16	R	C2
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GTCATATG	IGVFSM1633EQRW	033_AJ_10M_10	16	R	C6
ACTTTACC	IGVFSM1633EQRW	033_AJ_10M_10	16	T	C6
GTCATATG	IGVFSM0075HJSS	033_AJ_10M_11	16	R	C6
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GTGCTTCC	IGVFSM1911LHOP	032_AJ_10F_10	16	R	C7
CTTCTAAC	IGVFSM1911LHOP	032_AJ_10F_10	16	T	C7
GTGCTTCC	IGVFSM2604ZWZC	032_AJ_10F_11	16	R	C7
CTTCTAAC	IGVFSM2604ZWZC	032_AJ_10F_11	16	T	C7
ATGTGTTG	IGVFSM2230WLPO	035_AJ_10M_10	16	R	C8
CTATTTCA	IGVFSM2230WLPO	035_AJ_10M_10	16	T	C8
ATGTGTTG	IGVFSM7820FOJS	035_AJ_10M_11	16	R	C8
CTATTTCA	IGVFSM7820FOJS	035_AJ_10M_11	16	T	C8
CCATCTTG	IGVFSM5489EQDV	026_AJ_10F_08	16	R	C9
TCTCATGC	IGVFSM5489EQDV	026_AJ_10F_08	16	T	C9
CCATCTTG	IGVFSM5120VDBH	076_PWKJ_10F_08	16	R	C9
TCTCATGC	IGVFSM5120VDBH	076_PWKJ_10F_08	16	T	C9
TACTGTCT	IGVFSM9075IJZC	029_AJ_10M_08	16	R	C10
ATCCTTAC	IGVFSM9075IJZC	029_AJ_10M_08	16	T	C10
TACTGTCT	IGVFSM2744HFBR	077_PWKJ_10M_08	16	R	C10
ATCCTTAC	IGVFSM2744HFBR	077_PWKJ_10M_08	16	T	C10
TTCATCGC	IGVFSM6339UEAO	028_AJ_10F_08	16	R	C11
TAAATATC	IGVFSM6339UEAO	028_AJ_10F_08	16	T	C11
TTCATCGC	IGVFSM9958NNHY	078_PWKJ_10F_08	16	R	C11
TAAATATC	IGVFSM9958NNHY	078_PWKJ_10F_08	16	T	C11
ACTGTGGG	IGVFSM4824RRCE	031_AJ_10M_08	16	R	C12
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ACTGTGGG	IGVFSM0701NELT	079_PWKJ_10M_08	16	R	C12
TTACCTGC	IGVFSM0701NELT	079_PWKJ_10M_08	16	T	C12
TCTGTGCC	IGVFSM3184ILFA	076_PWKJ_10F_10	16	R	D1
CACTTTCA	IGVFSM3184ILFA	076_PWKJ_10F_10	16	T	D1
TCTGTGCC	IGVFSM0096BMWC	076_PWKJ_10F_11	16	R	D1
CACTTTCA	IGVFSM0096BMWC	076_PWKJ_10F_11	16	T	D1
TCAATCTC	IGVFSM4307KCOE	077_PWKJ_10M_10	16	R	D2
CACCTTTA	IGVFSM4307KCOE	077_PWKJ_10M_10	16	T	D2

TCAATCTC	IGVFSM5586TLOK	077_PWKJ_10M_11	16	R	D2
CACCTTTA	IGVFSM5586TLOK	077_PWKJ_10M_11	16	T	D2
GTCCTCTG	IGVFSM5409UMED	078_PWKJ_10F_10	16	R	D3
CTGACTTC	IGVFSM5409UMED	078_PWKJ_10F_10	16	T	D3
GTCCTCTG	IGVFSM1096FNGC	078_PWKJ_10F_11	16	R	D3
CTGACTTC	IGVFSM1096FNGC	078_PWKJ_10F_11	16	T	D3
TTACATTC	IGVFSM3239STTL	079_PWKJ_10M_10	16	R	D4
CATTTGGA	IGVFSM3239STTL	079_PWKJ_10M_10	16	T	D4
TTACATTC	IGVFSM2227UKRP	079_PWKJ_10M_11	16	R	D4
CATTTGGA	IGVFSM2227UKRP	079_PWKJ_10M_11	16	T	D4
ATTCTGTC	IGVFSM2642TMHC	080_PWKJ_10F_10	16	R	D5
GCTCTACT	IGVFSM2642TMHC	080_PWKJ_10F_10	16	T	D5
ATTCTGTC	IGVFSM6638YNDH	080_PWKJ_10F_11	16	R	D5
GCTCTACT	IGVFSM6638YNDH	080_PWKJ_10F_11	16	T	D5
TGTGTATG	IGVFSM3142HYPK	081_PWKJ_10M_10	16	R	D6
GTTACGTA	IGVFSM3142HYPK	081_PWKJ_10M_10	16	T	D6
TGTGTATG	IGVFSM1277GJRQ	081_PWKJ_10M_11	16	R	D6
GTTACGTA	IGVFSM1277GJRQ	081_PWKJ_10M_11	16	T	D6
TCCATTTG	IGVFSM6801NQEI	084_PWKJ_10F_10	16	R	D7
CCTGTTGC	IGVFSM6801NQEI	084_PWKJ_10F_10	16	T	D7
TCCATTTG	IGVFSM9469AVQG	084_PWKJ_10F_11	16	R	D7
CCTGTTGC	IGVFSM9469AVQG	084_PWKJ_10F_11	16	T	D7
TTAGCTTC	IGVFSM5592JXHF	085_PWKJ_10M_10	16	R	D8
CTATCATC	IGVFSM5592JXHF	085_PWKJ_10M_10	16	T	D8
TTAGCTTC	IGVFSM1980ATRP	085_PWKJ_10M_11	16	R	D8
CTATCATC	IGVFSM1980ATRP	085_PWKJ_10M_11	16	T	D8
GTGCTTGA	IGVFSM7056LBHR	030_AJ_10F_08	16	R	D9
GCTATCAT	IGVFSM7056LBHR	030_AJ_10F_08	16	T	D9
GTGCTTGA	IGVFSM1448KEUX	080_PWKJ_10F_08	16	R	D9
GCTATCAT	IGVFSM1448KEUX	080_PWKJ_10F_08	16	T	D9
GTTTGTGA	IGVFSM6013ZWZO	033_AJ_10M_08	16	R	D10
ACATTCAT	IGVFSM6013ZWZO	033_AJ_10M_08	16	T	D10
GTTTGTGA	IGVFSM8451AOOU	081_PWKJ_10M_08	16	R	D10
ACATTCAT	IGVFSM8451AOOU	081_PWKJ_10M_08	16	T	D10
GAAATTAG	IGVFSM8570TOBS	032_AJ_10F_08	16	R	D11

TTCGCTAC	IGVFSM8570TOBS	032_AJ_10F_08	16	T	D11
GAAATTAG	IGVFSM4227UNYC	084_PWKJ_10F_08	16	R	D11
TTCGCTAC	IGVFSM4227UNYC	084_PWKJ_10F_08	16	T	D11
GCAAATTC	IGVFSM4324XCGU	035_AJ_10M_08	16	R	D12
CATTCTAC	IGVFSM4324XCGU	035_AJ_10M_08	16	T	D12
GCAAATTC	IGVFSM1696GVKP	085_PWKJ_10M_08	16	R	D12
CATTCTAC	IGVFSM1696GVKP	085_PWKJ_10M_08	16	T	D12
GAGGTTGA	IGVFSM6816QWSB	036_129S1J_10F_10	16	R	E1
CACTTATC	IGVFSM6816QWSB	036_129S1J_10F_10	16	T	E1
GAGGTTGA	IGVFSM8822PZTL	036_129S1J_10F_11	16	R	E1
CACTTATC	IGVFSM8822PZTL	036_129S1J_10F_11	16	T	E1
CCTGTCTG	IGVFSM1094VAAI	037_129S1J_10M_10	16	R	E2
ATAAGCTC	IGVFSM1094VAAI	037_129S1J_10M_10	16	T	E2
CCTGTCTG	IGVFSM9038UIJU	037_129S1J_10M_11	16	R	E2
ATAAGCTC	IGVFSM9038UIJU	037_129S1J_10M_11	16	T	E2
GTGGGTTC	IGVFSM2570ENQI	038_129S1J_10F_10	16	R	E3
TCATCCTG	IGVFSM2570ENQI	038_129S1J_10F_10	16	T	E3
GTGGGTTC	IGVFSM1263AYLV	038_129S1J_10F_11	16	R	E3
TCATCCTG	IGVFSM1263AYLV	038_129S1J_10F_11	16	T	E3
TTTGCATC	IGVFSM8496DJXG	041_129S1J_10M_10	16	R	E4
CCTGGTAT	IGVFSM8496DJXG	041_129S1J_10M_10	16	T	E4
TTTGCATC	IGVFSM9236NDAG	041_129S1J_10M_11	16	R	E4
CCTGGTAT	IGVFSM9236NDAG	041_129S1J_10M_11	16	T	E4
AGGTAATA	IGVFSM7330MCOB	040_129S1J_10F_10	16	R	E5
TGGTATAC	IGVFSM7330MCOB	040_129S1J_10F_10	16	T	E5
AGGTAATA	IGVFSM9813UIKP	040_129S1J_10F_11	16	R	E5
TGGTATAC	IGVFSM9813UIKP	040_129S1J_10F_11	16	T	E5
GTGCCTTC	IGVFSM6313PLBW	043_129S1J_10M_10	16	R	E6
TTGGGAGA	IGVFSM6313PLBW	043_129S1J_10M_10	16	T	E6
GTGCCTTC	IGVFSM9650VFSK	043_129S1J_10M_11	16	R	E6
TTGGGAGA	IGVFSM9650VFSK	043_129S1J_10M_11	16	T	E6
ATGTTTCC	IGVFSM4841JIOO	044_129S1J_10F_10	16	R	E7
ACTTCATC	IGVFSM4841JIOO	044_129S1J_10F_10	16	T	E7
ATGTTTCC	IGVFSM5851CIQU	044_129S1J_10F_11	16	R	E7
ACTTCATC	IGVFSM5851CIQU	044_129S1J_10F_11	16	T	E7

CTTAATTC	IGVFSM7262NIVF	045_129S1J_10M_10	16	R	E8
TCTCTAGC	IGVFSM7262NIVF	045_129S1J_10M_10	16	T	E8
CTTAATTC	IGVFSM0998LUIZ	045_129S1J_10M_11	16	R	E8
TCTCTAGC	IGVFSM0998LUIZ	045_129S1J_10M_11	16	T	E8
TCTGGCTC	IGVFSM1632QLUR	036_129S1J_10F_08	16	R	E9
ATGCCCTT	IGVFSM1632QLUR	036_129S1J_10F_08	16	T	E9
TCTGGCTC	IGVFSM2352QEZL	086_CASTJ_10F_08	16	R	E9
ATGCCCTT	IGVFSM2352QEZL	086_CASTJ_10F_08	16	T	E9
CATCATTT	IGVFSM9668NECE	037_129S1J_10M_08	16	R	E10
CCCAATTT	IGVFSM9668NECE	037_129S1J_10M_08	16	T	E10
CATCATTT	IGVFSM4914MZNP	087_CASTJ_10M_08	16	R	E10
CCCAATTT	IGVFSM4914MZNP	087_CASTJ_10M_08	16	T	E10
GTTGTCTC	IGVFSM0340EAYH	038_129S1J_10F_08	16	R	E11
ACTATATA	IGVFSM0340EAYH	038_129S1J_10F_08	16	T	E11
GTTGTCTC	IGVFSM2528JPFP	088_CASTJ_10F_08	16	R	E11
ACTATATA	IGVFSM2528JPFP	088_CASTJ_10F_08	16	T	E11
ATCTTCTG	IGVFSM4755QPGT	041_129S1J_10M_08	16	R	E12
CTCTATAC	IGVFSM4755QPGT	041_129S1J_10M_08	16	T	E12
ATCTTCTG	IGVFSM0740TEWR	089_CASTJ_10M_08	16	R	E12
CTCTATAC	IGVFSM0740TEWR	089_CASTJ_10M_08	16	T	E12
TGTTTGCC	IGVFSM7969ZAUZ	086_CASTJ_10F_10	16	R	F1
CTGTCTCA	IGVFSM7969ZAUZ	086_CASTJ_10F_10	16	T	F1
TGTTTGCC	IGVFSM8330OVCV	086_CASTJ_10F_11	16	R	F1
CTGTCTCA	IGVFSM8330OVCV	086_CASTJ_10F_11	16	T	F1
TTCTGTCA	IGVFSM3300FHLI	087_CASTJ_10M_10	16	R	F2
GACCTTTC	IGVFSM3300FHLI	087_CASTJ_10M_10	16	T	F2
TTCTGTCA	IGVFSM4100ZICM	087_CASTJ_10M_11	16	R	F2
GACCTTTC	IGVFSM4100ZICM	087_CASTJ_10M_11	16	T	F2
ACGGACTC	IGVFSM0900XHGX	088_CASTJ_10F_10	16	R	F3
GATTTGGC	IGVFSM0900XHGX	088_CASTJ_10F_10	16	T	F3
ACGGACTC	IGVFSM2098KNMX	088_CASTJ_10F_11	16	R	F3
GATTTGGC	IGVFSM2098KNMX	088_CASTJ_10F_11	16	T	F3
TTTGGTCA	IGVFSM7381JEDM	089_CASTJ_10M_10	16	R	F4
CGTCTAGG	IGVFSM7381JEDM	089_CASTJ_10M_10	16	T	F4
TTTGGTCA	IGVFSM9723WBIN	089_CASTJ_10M_11	16	R	F4

CGTCTAGG	IGVFSM9723WBIN	089_CASTJ_10M_11	16	T	F4
TATCCGGG	IGVFSM5920YOJY	090_CASTJ_10F_10	16	R	F5
TACTCGAA	IGVFSM5920YOJY	090_CASTJ_10F_10	16	T	F5
TATCCGGG	IGVFSM1103DUUT	090_CASTJ_10F_11	16	R	F5
TACTCGAA	IGVFSM1103DUUT	090_CASTJ_10F_11	16	T	F5
TGTCATTC	IGVFSM0628FIYW	091_CASTJ_10M_10	16	R	F6
CAGCCTTT	IGVFSM0628FIYW	091_CASTJ_10M_10	16	T	F6
TGTCATTC	IGVFSM6763UWGN	091_CASTJ_10M_11	16	R	F6
CAGCCTTT	IGVFSM6763UWGN	091_CASTJ_10M_11	16	T	F6
ATTCTCTG	IGVFSM6711SYYK	094_CASTJ_10F_10	16	R	F7
CCTCATTA	IGVFSM6711SYYK	094_CASTJ_10F_10	16	T	F7
ATTCTCTG	IGVFSM8588WXUX	094_CASTJ_10F_11	16	R	F7
CCTCATTA	IGVFSM8588WXUX	094_CASTJ_10F_11	16	T	F7
TGGCTTCC	IGVFSM9677NTDJ	093_CASTJ_10M_10	16	R	F8
CTTATACC	IGVFSM9677NTDJ	093_CASTJ_10M_10	16	T	F8
TGGCTTCC	IGVFSM7066GBZS	093_CASTJ_10M_11	16	R	F8
CTTATACC	IGVFSM7066GBZS	093_CASTJ_10M_11	16	T	F8
TTGTTGCC	IGVFSM9964SESF	040_129S1J_10F_08	16	R	F9
TCTATTAC	IGVFSM9964SESF	040_129S1J_10F_08	16	T	F9
TTGTTGCC	IGVFSM8717VBUZ	090_CASTJ_10F_08	16	R	F9
TCTATTAC	IGVFSM8717VBUZ	090_CASTJ_10F_08	16	T	F9
GTCATCTC	IGVFSM4576HVJC	043_129S1J_10M_08	16	R	F10
CCTGCATT	IGVFSM4576HVJC	043_129S1J_10M_08	16	T	F10
GTCATCTC	IGVFSM5442GAOY	091_CASTJ_10M_08	16	R	F10
CCTGCATT	IGVFSM5442GAOY	091_CASTJ_10M_08	16	T	F10
TTGCTCAT	IGVFSM3033SSIU	044_129S1J_10F_08	16	R	F11
CAATCCTT	IGVFSM3033SSIU	044_129S1J_10F_08	16	T	F11
TTGCTCAT	IGVFSM4807XVYB	094_CASTJ_10F_08	16	R	F11
CAATCCTT	IGVFSM4807XVYB	094_CASTJ_10F_08	16	T	F11
CTGTCTGC	IGVFSM3513BNRI	045_129S1J_10M_08	16	R	F12
TTGTCTTA	IGVFSM3513BNRI	045_129S1J_10M_08	16	T	F12
CTGTCTGC	IGVFSM5275FTRH	095_CASTJ_10M_08	16	R	F12
TTGTCTTA	IGVFSM5275FTRH	095_CASTJ_10M_08	16	T	F12
TATATTCC	IGVFSM8653IGLV	058_WSBJ_10F_10	16	R	G1
TCACTTTA	IGVFSM8653IGLV	058_WSBJ_10F_10	16	T	G1

TATATTCC	IGVFSM7260RGBF	058_WSBJ_10F_11	16	R	G1
TCACTTTA	IGVFSM7260RGBF	058_WSBJ_10F_11	16	T	G1
ATATTGGC	IGVFSM6049JASP	057_WSBJ_10M_10	16	R	G2
TGCTTGGG	IGVFSM6049JASP	057_WSBJ_10M_10	16	T	G2
ATATTGGC	IGVFSM8555WNHX	057_WSBJ_10M_11	16	R	G2
TGCTTGGG	IGVFSM8555WNHX	057_WSBJ_10M_11	16	T	G2
GTGTCCTC	IGVFSM1827LDDT	060_WSBJ_10F_10	16	R	G3
CGCTCATT	IGVFSM1827LDDT	060_WSBJ_10F_10	16	T	G3
GTGTCCTC	IGVFSM5070ERGT	060_WSBJ_10F_11	16	R	G3
CGCTCATT	IGVFSM5070ERGT	060_WSBJ_10F_11	16	T	G3
ATCTTCAT	IGVFSM0997ZFEX	059_WSBJ_10M_10	16	R	G4
GCCTCTAT	IGVFSM0997ZFEX	059_WSBJ_10M_10	16	T	G4
ATCTTCAT	IGVFSM6693YSWV	059_WSBJ_10M_11	16	R	G4
GCCTCTAT	IGVFSM6693YSWV	059_WSBJ_10M_11	16	T	G4
CGTGGTTG	IGVFSM7348XFYJ	062_WSBJ_10F_10	16	R	G5
GAGCACAA	IGVFSM7348XFYJ	062_WSBJ_10F_10	16	T	G5
CGTGGTTG	IGVFSM0359CZGZ	062_WSBJ_10F_11	16	R	G5
GAGCACAA	IGVFSM0359CZGZ	062_WSBJ_10F_11	16	T	G5
TTGCATCC	IGVFSM4977BSZT	061_WSBJ_10M_10	16	R	G6
CTCTTAAC	IGVFSM4977BSZT	061_WSBJ_10M_10	16	T	G6
TTGCATCC	IGVFSM1160IHPD	061_WSBJ_10M_11	16	R	G6
CTCTTAAC	IGVFSM1160IHPD	061_WSBJ_10M_11	16	T	G6
TCTTAATC	IGVFSM2781SEWP	096_WSBJ_10F_10	16	R	G7
TCTAGGCT	IGVFSM2781SEWP	096_WSBJ_10F_10	16	T	G7
TCTTAATC	IGVFSM0153ZUFQ	096_WSBJ_10F_11	16	R	G7
TCTAGGCT	IGVFSM0153ZUFQ	096_WSBJ_10F_11	16	T	G7
TGCATTTC	IGVFSM4132OPDD	063_WSBJ_10M_10	16	R	G8
AATTCTGC	IGVFSM4132OPDD	063_WSBJ_10M_10	16	T	G8
TGCATTTC	IGVFSM9229AVFG	063_WSBJ_10M_11	16	R	G8
AATTCTGC	IGVFSM9229AVFG	063_WSBJ_10M_11	16	T	G8
GATGTTTC	IGVFSM9456XQYT	046_NZOJ_10F_08	16	R	G9
CATTCTCA	IGVFSM9456XQYT	046_NZOJ_10F_08	16	T	G9
GATGTTTC	IGVFSM0232VKZB	058_WSBJ_10F_08	16	R	G9
CATTCTCA	IGVFSM0232VKZB	058_WSBJ_10F_08	16	T	G9
ATCTTGTC	IGVFSM8242IPDC	047_NZOJ_10M_08	16	R	G10

ACTTGCCT	IGVFSM8242IPDC	047_NZOJ_10M_08	16	T	G10
ATCTTGTC	IGVFSM9417BLQR	057_WSBJ_10M_08	16	R	G10
ACTTGCCT	IGVFSM9417BLQR	057_WSBJ_10M_08	16	T	G10
TCATATTC	IGVFSM4848LHPG	048_NZOJ_10F_08	16	R	G11
ATCATTGC	IGVFSM4848LHPG	048_NZOJ_10F_08	16	T	G11
TCATATTC	IGVFSM0760BMNA	060_WSBJ_10F_08	16	R	G11
ATCATTGC	IGVFSM0760BMNA	060_WSBJ_10F_08	16	T	G11
TGGCCTCT	IGVFSM3199ZHGC	049_NZOJ_10M_08	16	R	G12
GTTCAACA	IGVFSM3199ZHGC	049_NZOJ_10M_08	16	T	G12
TGGCCTCT	IGVFSM4075DZPE	059_WSBJ_10M_08	16	R	G12
GTTCAACA	IGVFSM4075DZPE	059_WSBJ_10M_08	16	T	G12
CGTTGTCT	IGVFSM6124VCIY	046_NZOJ_10F_10	16	R	H1
CCATTTGC	IGVFSM6124VCIY	046_NZOJ_10F_10	16	T	H1
CGTTGTCT	IGVFSM8844CIBF	046_NZOJ_10F_11	16	R	H1
CCATTTGC	IGVFSM8844CIBF	046_NZOJ_10F_11	16	T	H1
TCTTGTCA	IGVFSM5941JXID	047_NZOJ_10M_10	16	R	H2
GACTTTGC	IGVFSM5941JXID	047_NZOJ_10M_10	16	T	H2
TCTTGTCA	IGVFSM2925KIGE	047_NZOJ_10M_11	16	R	H2
GACTTTGC	IGVFSM2925KIGE	047_NZOJ_10M_11	16	T	H2
TATTCCTG	IGVFSM3735WYFG	048_NZOJ_10F_10	16	R	H3
ATTGGCTC	IGVFSM3735WYFG	048_NZOJ_10F_10	16	T	H3
TATTCCTG	IGVFSM7325IMAE	048_NZOJ_10F_11	16	R	H3
ATTGGCTC	IGVFSM7325IMAE	048_NZOJ_10F_11	16	T	H3
TCCATGTC	IGVFSM5754TPRZ	049_NZOJ_10M_10	16	R	H4
GTGCTAGC	IGVFSM5754TPRZ	049_NZOJ_10M_10	16	T	H4
TCCATGTC	IGVFSM6425UPLJ	049_NZOJ_10M_11	16	R	H4
GTGCTAGC	IGVFSM6425UPLJ	049_NZOJ_10M_11	16	T	H4
TTGTCATC	IGVFSM1713ZBFY	050_NZOJ_10F_10	16	R	H5
CTTTCAAC	IGVFSM1713ZBFY	050_NZOJ_10F_10	16	T	H5
TTGTCATC	IGVFSM4889EESW	050_NZOJ_10F_11	16	R	H5
CTTTCAAC	IGVFSM4889EESW	050_NZOJ_10F_11	16	T	H5
ATTTCCTG	IGVFSM2896IHAT	051_NZOJ_10M_10	16	R	H6
ACTATTGC	IGVFSM2896IHAT	051_NZOJ_10M_10	16	T	H6
ATTTCCTG	IGVFSM0427HINK	051_NZOJ_10M_11	16	R	H6
ACTATTGC	IGVFSM0427HINK	051_NZOJ_10M_11	16	T	H6

GTGTCTCC	IGVFSM8339CJUO	052_NZOJ_10F_10	16	R	H7
ACTGGCTT	IGVFSM8339CJUO	052_NZOJ_10F_10	16	T	H7
GTGTCTCC	IGVFSM3012TVCO	052_NZOJ_10F_11	16	R	H7
ACTGGCTT	IGVFSM3012TVCO	052_NZOJ_10F_11	16	T	H7
GTGTGTGT	IGVFSM2662INEM	053_NZOJ_10M_10	16	R	H8
ATTAGGCT	IGVFSM2662INEM	053_NZOJ_10M_10	16	T	H8
GTGTGTGT	IGVFSM7621BJTR	053_NZOJ_10M_11	16	R	H8
ATTAGGCT	IGVFSM7621BJTR	053_NZOJ_10M_11	16	T	H8
TATGCTTC	IGVFSM4326PPHF	050_NZOJ_10F_08	16	R	H9
GCCTTTCA	IGVFSM4326PPHF	050_NZOJ_10F_08	16	T	H9
TATGCTTC	IGVFSM4265ZXJM	062_WSBJ_10F_08	16	R	H9
GCCTTTCA	IGVFSM4265ZXJM	062_WSBJ_10F_08	16	T	H9
ATGGTGTT	IGVFSM7580WBSP	051_NZOJ_10M_08	16	R	H10
ATTCTAGG	IGVFSM7580WBSP	051_NZOJ_10M_08	16	T	H10
ATGGTGTT	IGVFSM8686RRRP	061_WSBJ_10M_08	16	R	H10
ATTCTAGG	IGVFSM8686RRRP	061_WSBJ_10M_08	16	T	H10
GAATAATG	IGVFSM0525KCUV	052_NZOJ_10F_08	16	R	H11
CCTTACAT	IGVFSM0525KCUV	052_NZOJ_10F_08	16	T	H11
GAATAATG	IGVFSM3681WOHF	096_WSBJ_10F_08	16	R	H11
CCTTACAT	IGVFSM3681WOHF	096_WSBJ_10F_08	16	T	H11
CCTCTGTG	IGVFSM5808CHQB	053_NZOJ_10M_08	16	R	H12
ACATTTGG	IGVFSM5808CHQB	053_NZOJ_10M_08	16	T	H12
CCTCTGTG	IGVFSM7664LZLU	063_WSBJ_10M_08	16	R	H12
ACATTTGG	IGVFSM7664LZLU	063_WSBJ_10M_08	16	T	H12

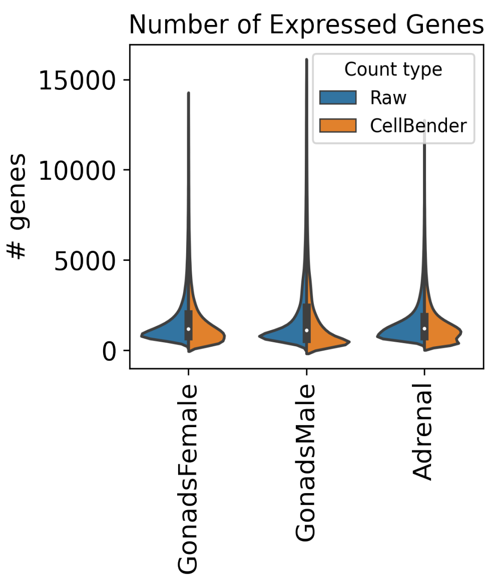
6. Description of data processing workflow and h5ad files

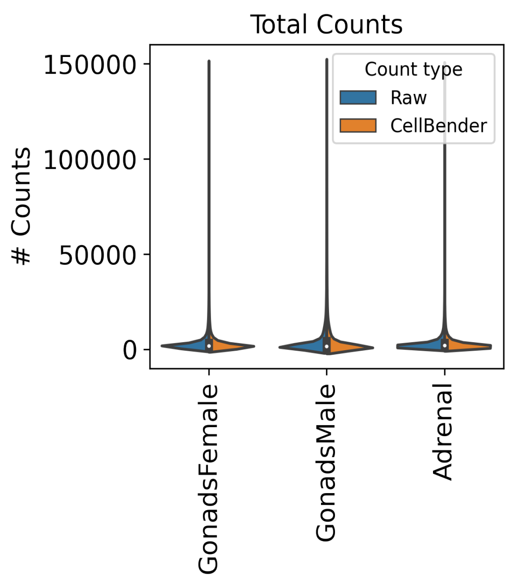
Data from all 9 subpools were concatenated into one AnnData object per tissue and combined with matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

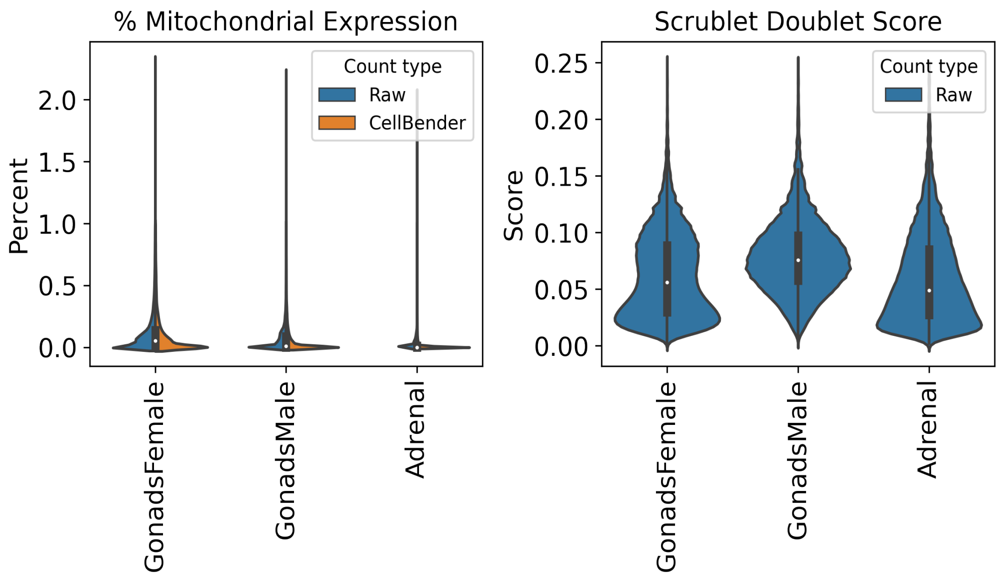
In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

Table 8: QC thresholds

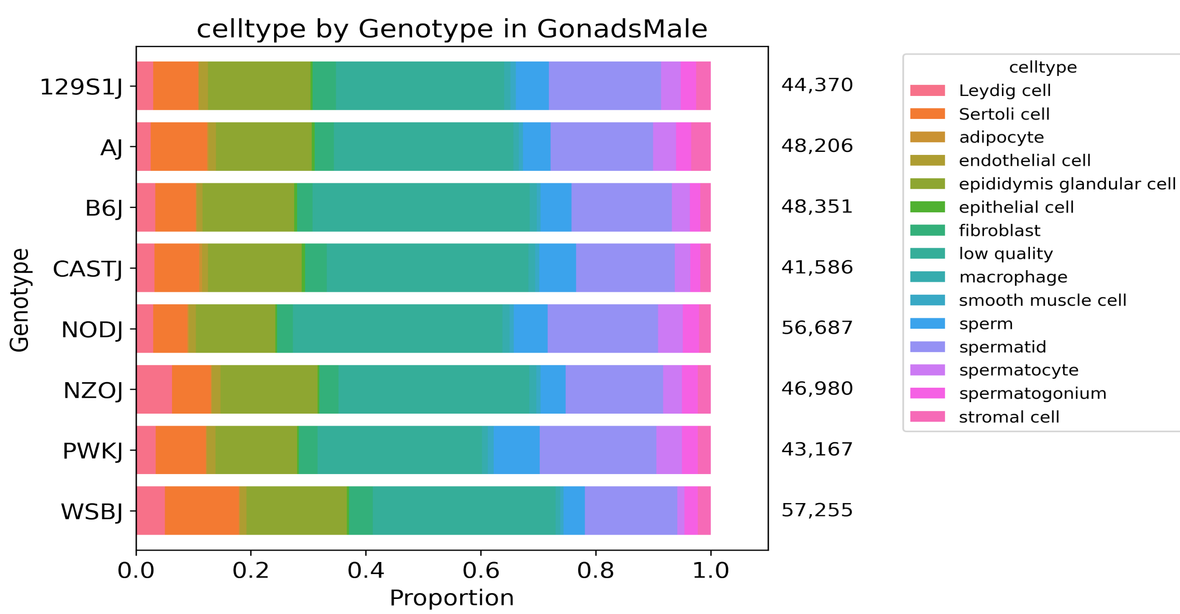
Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25

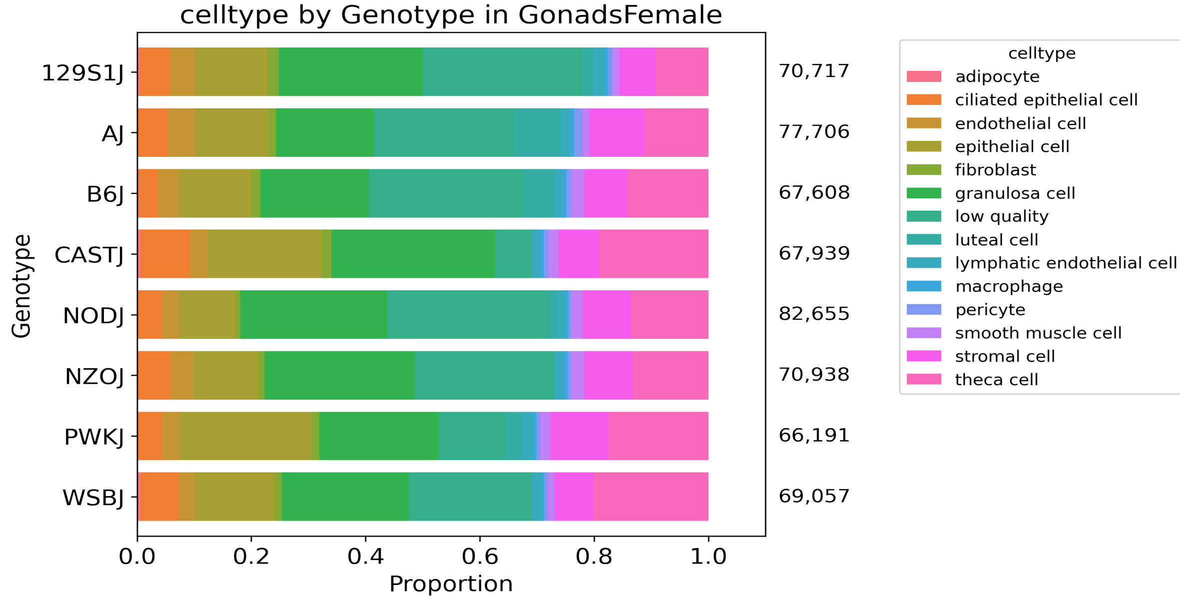






After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 45 total clusters in GonadsMale, 40 total clusters in GonadsFemale, and 41 clusters in Adrenal, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.





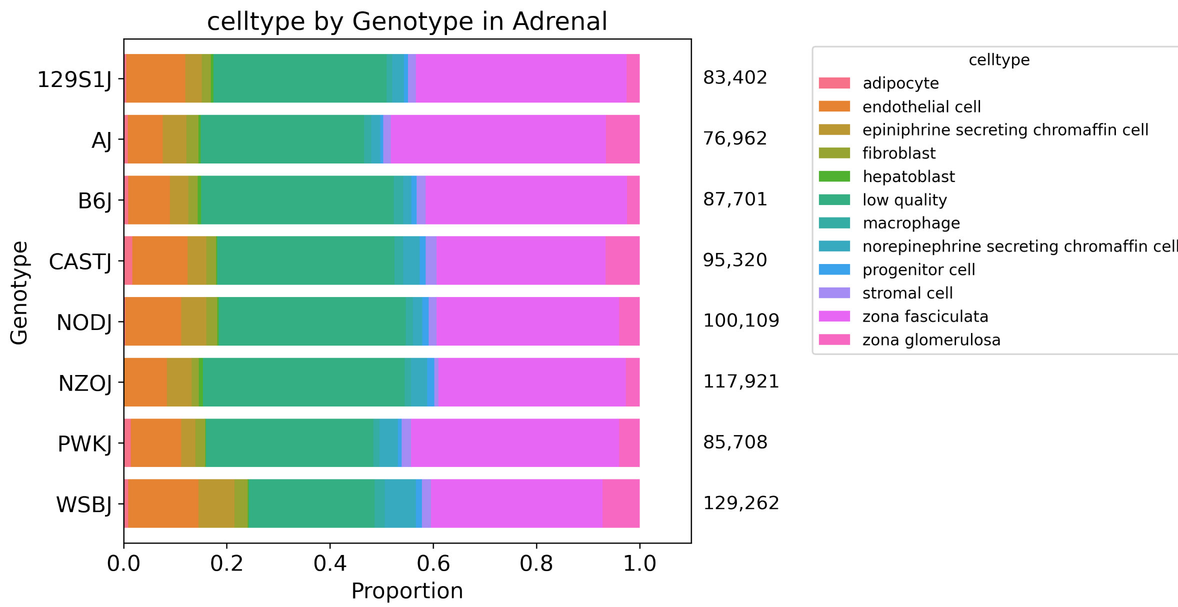


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.

SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.
Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).

Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.
cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0

cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Plate_report_igvf_008.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/GonadsMale.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Adrenal.ipynb